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PC-42 through PC-52**PC-01 - Moderated****Progressive Mild Leaflet Tethering is a Normal Developmental Feature of Tricuspid Valve Maturation During Youth: A Quantitative Three-Dimensional Echocardiography (3DE) Study**

Martha Maria Moran¹, Tuqa Al Lawati¹, Nee Sze Khoo¹, Timothy Colen¹, Justin Grenier², Richard Thompson¹, Lily Lin¹. ¹Stollery Children Hospital, Edmonton, AB, Canada; ²University of Alberta, Edmonton, AB, Canada.

Background: Recent study on mitral valve development in healthy children suggest developmental differences between leaflets, with greater relative expansion of the anterior leaflet while maintaining annular saddle shape. To our knowledge, there is no published literature on developmental changes of the tricuspid valve (TV). Using 3DE we aim to quantify developmental changes of the TV annulus and leaflets. **Methods:** This cross-sectional study used prospectively acquired TV 3DE volumetric datasets in 32 healthy children between 0 and 18 years of age. Using a custom designed TV MATLAB software, annular dimensions, sphericity index (lateral width/anteroposterior annular dimension ratio) and bending angle were analyzed in mid-systole. TV leaflets were segmented into anterior (AL), septal (SL) and posterior (PL) to measure regional areas, prolapse and tethering volumes. Subjects were divided into 3 age groups: 0-5, 6-12 and 13-18 years. Between group comparison was performed using the Kruskal-Wallis test with significance at $p < 0.05$. Values expressed as mean (standard deviation). **Results:** Mean age in months was 30.4 (22.9) for Group 1, 114.5 (22.4) for Group 2 and 182.2 (17.7) for Group 3. TV annulus and leaflet parameters were detailed in Table 1. With increasing age and size, TV annulus enlarged in both anteroposterior and lateral dimensions symmetrically such that annular sphericity index remained unchanged. The "saddle-shape" of the annulus, quantified through the annular bending angle, was also maintained through groups. There was size proportionate expansion of TV leaflets with increasing age (stable total leaflet area indexed by BSA). Individual leaflets expanded symmetrically. With increasing age, there was no significant change in leaflet prolapse ($p=0.1$) but an increase in leaflet tethering ($p=0.01$). This progressive leaflet tethering was slightly more pronounced in SL and AL over PL. **Conclusion:** Our novel study suggest the normal TV maturation involved preservation of annular geometry and symmetric leaflets expansion, however there is mild progressive leaflet tethering. Whether this leaflet tethering is the result of physiologic differences in growth potential between leaflet and subvalve tensor apparatus or an adaptive biologic stimulus for leaflet growth, warrants further exploration.

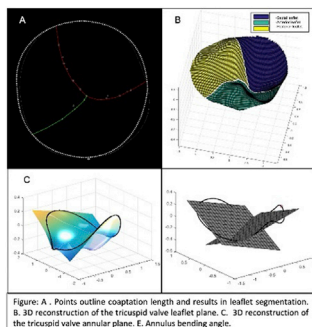


Table: Comparisons of 3DE TV annular and leaflet parameters between the 3 age groups. NS denotes non-significance.

variables	Group 1: 0-5 (N=8)	Group 2: 6-12 (N=14)	Group 3: 13-18 (N=10)	p-value
Age (months)	30.4 (22.9)	114.5 (22.4)	182.2 (17.7)	
Body surface area (BSA) (m ²)	0.87 (0.3)	1.15 (0.3)	1.73 (0.2)	
Annular parameters				
TV annulus anteroposterior dimension (cm)	1.6 (0.3)	2.1 (0.2)	2.5 (0.3)	<0.001
TV annulus lateral dimension (cm)	1.6 (0.2)	2.1 (0.2)	2.5 (0.3)	<0.001
TV sphericity index	1.2 (0.1)	1.2 (0.1)	1.3 (0.1)	NS
Bending angle (deg/cm)	105.2 (8.7)	103.6 (11.1)	101.6 (10.2)	NS
Leaflet parameters				
Total leaflet area (cm ²)	2.6 (1.3)	6.1 (2.1)	8.9 (3.1)	<0.001
Total leaflet area indexed by BSA (cm ² /m ²)	4.7 (1.7)	4.4 (0.8)	4.1 (1.4)	NS
AL total leaflet area (cm ²)	0.3 (0.05)	0.4 (0.20)	0.4 (0.3)	NS
SL total leaflet area (cm ²)	0.4 (0.07)	0.3 (0.07)	0.4 (0.08)	NS
PL total leaflet area (cm ²)	0.3 (0.1)	0.3 (0.08)	0.3 (0.1)	NS
Total leaflet prolapse volume indexed by BSA (cm ³ /m ²)	0.1 (0.2)	0.02 (0.02)	0.08 (0.2)	NS
Total leaflet tethering volume indexed by BSA (cm ³ /m ²)	0.3 (0.3)	0.4 (0.2)	0.7 (0.4)	0.01
AL tethering volume indexed by BSA (cm ³ /m ²)	0.08 (0.20)	0.1 (0.08)	0.3 (0.3)	0.03
SL tethering volume indexed by BSA (cm ³ /m ²)	0.08 (0.20)	0.2 (0.08)	0.3 (0.2)	0.02
PL tethering volume indexed by BSA (cm ³ /m ²)	0.08 (0.1)	0.1 (0.05)	0.2 (0.1)	0.06

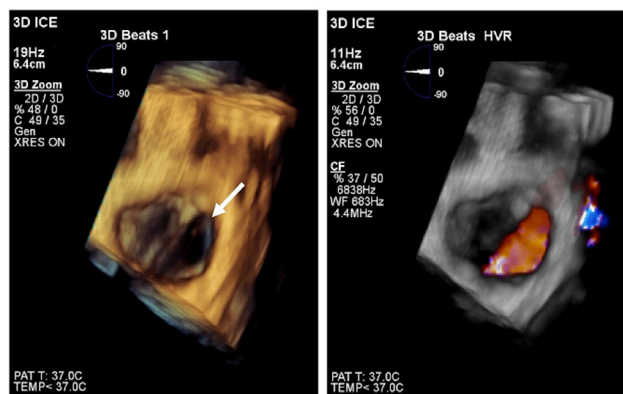
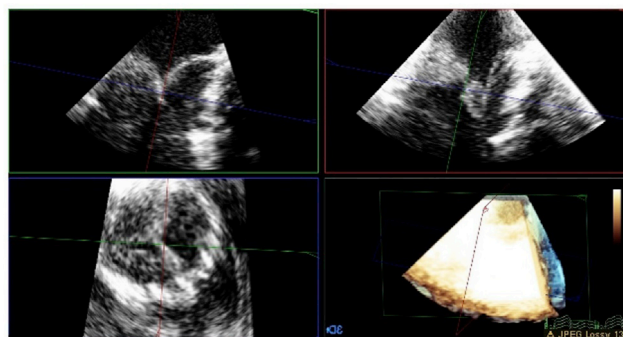
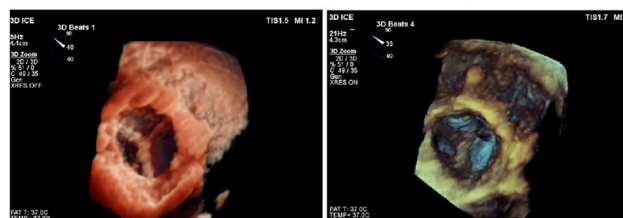
Figure 1: A. Points outline coaptation length and results in leaflet segmentation. B. 3D reconstruction of the tricuspid valve leaflet plane. C. 3D reconstruction of the tricuspid valve annular plane. D. Annulus bending angle.

PC-02 - Oral**Three-Dimensional Intracardiac Echocardiography in Transcatheter Pulmonary Valve Replacement**

Cesar Gonzalez de Alba, Jenny Zablah, Gareth Morgan, Dale Burkett, Pei-Ni Jone. Children's Hospital Colorado - University of Colorado, Aurora, CO.

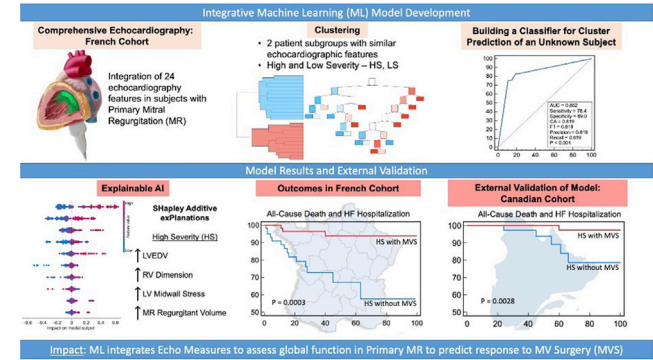
Background: Intracardiac echocardiography (ICE) allows real-time evaluation of anatomical structures during interventional procedures for structural heart disease. ICE has been shown to decrease patient and operator exposure to fluoroscopy during transcatheter inter-

ventional procedures. Three-dimensional (3D) ICE is a new technique that allows improved visualization and assessment of cardiac structures and guide interventional procedures. Use of 3D-ICE has not been previously described for transcatheter pulmonary valve replacement (PVR). **Methods:** Single center, retrospective case review of young adults with congenital heart disease who underwent 3D-ICE after transcatheter PVR from 12/1/2021 - 02/15/2022. Extracted clinical data included patient demographics, ICE images, transthoracic echo (TTE) images and available follow up data. **Results:** Eight patients underwent ICE immediately after transcatheter PVR. Median age was 20 years. Tetralogy of Fallot was the most common baseline diagnosis. On a qualitative scale, 2D-ICE images of the pulmonary valve were excellent (4/4) for all patients. Three-dimensional ICE (Figure 1) with multiplane reconstruction (Figure 2) were very good (3.6/4). A TTE performed the day after the procedure showed lower quality visualization of the valve (2.4/4). While most patients had no significant abnormalities on ICE, one patient was found to have a partially immobile leaflet that was best seen by 3D-ICE (Figure 3). No patient required re-intervention. **Conclusion:** 3D-ICE after transcatheter PVR is feasible and allows for improved visualization of all valve leaflets and immediate identification of complications that may be difficult to recognize by other echocardiographic modalities.

**PC-03 - Oral****Integrating Echocardiography Parameters with Explainable Artificial Intelligence for Data-Driven Clustering of Primary Mitral Regurgitation Severity Phenotypes**

Jérémy Bernard¹, Naveena Yanamala², Rohan Shah³, Karthik Seetharam², Alexandre Altes³, Marlene Dupuis¹, Oumhani Toubal¹, Haöfa Mahjoub¹, Hélène Dumortier³, Jean Tartar³, Erwan Salaun¹, Kim O'Connor⁴, Mathieu Bernier¹, Jonathan Beaudoin¹, Nancy Côté², André Vincentelli¹, Florent LeVen⁵, Sylvestre Maréchaux³, Philippe Pibarot¹, Partho P. Sengupta². ¹Institut Universitaire de Cardiologie et de Pneumologie de Québec - Université Laval, Québec, QC, Canada; ²Robert Wood Johnson University Hospital (RWJUH) and Rutgers Robert Wood Johnson Medical School (RWJMS), New Brunswick, NJ; ³Department of Cardiology, GCS-Groupement des hôpitaux de l'Institut Catholique de Lille - Université Catholique de Lille, Lille, France; ⁴Cardiac Surgery Department, Centre Hospitalier Régional et Universitaire de Lille, Lille, France; ⁵Department of Cardiology, Hôpital La Cavale Blanche - Centre Hospitalier Régional Universitaire de Brest, Brest, France.

Background: Primary mitral regurgitation (MR) is a heterogeneous disease where echocardiography parameters are integrated using guideline-driven recommendations to identify disease severity and help management. Our objective was to explore novel data-driven approaches using machine learning (ML) and explainable artificial intelligence (AI) to delineate phenotypes of primary MR that benefit from mitral valve surgery (either repair/replacement). **Methods:** We explored 400 subjects from two prospective cohorts with primary MR followed up in France (n=243; development cohort) and Canada (n=157; validation cohort). Unsupervised and supervised ML and explainable AI were used to integrate 24 standard and commonly used echo parameters to identify echo-derived phenotypes of MR severity (Figure). We compared the phenogroups' incremental prognostic value over conventional MR profiles and for the primary endpoint of all-cause mortality and heart failure hospitalization, stratified by the occurrence of surgery. **Results:** In the France Cohort, we identified a High Severity (HS) phenogroup (n=118) who had improved event-free survival following surgery ($p<0.001$) versus those without surgery, and a Low Severity (LS) phenogroup (n=125) without significant event-free survival improvement irrespective of surgery occurrence ($p=0.89$). The validation Canadian cohort confirmed these observations for both HS ($p=0.003$, n=90) and LS ($p=0.06$, n=67) phenogroups. The echo-derived phenogrouping provided high predictive performance (AUC=0.85, 78% sensitivity and 89% specificity, $p<0.001$). For patients conventionally categorized to have moderate-to-severe or severe MR, the phenogrouping approach presented incremental prognostic value at 9 years (Harrell's C-statistic [0.71 vs. 0.75 $p=0.03$], net reclassification improvement [0.75, 95%CI 0.31-1.19, $p<0.001$], and integrated discrimination improvement [0.040, 95%CI 0.005-0.075, $p=0.02$]). Explainable AI specified how and the degree to which each echocardiographic parameter contributed to phenogroup distribution (Figure). **Conclusions:** Novel data-driven phenogrouping approaches using explainable AI aids the integration of echocardiographic data for identifying patients with primary MR who have improved event-free survival following surgery.

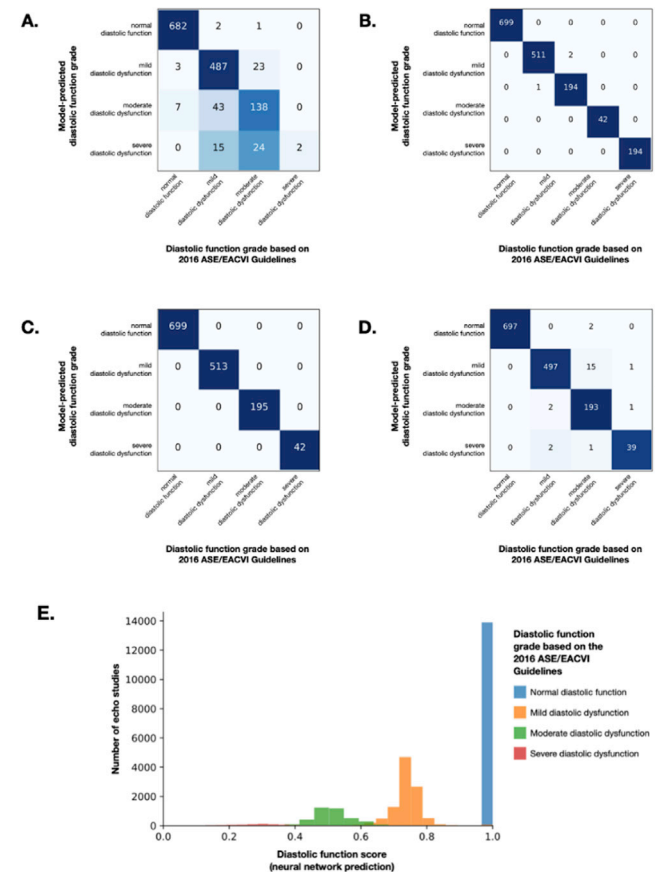


PC-04 - Oral

A Novel Continuous Left Ventricular Diastolic Function Score Using Machine Learning
River Jiang, Darwin Yeung, Delaram Behnami, Christina Luong, Michael Tsang, John Jue, Ken Gin, Parvathy Nair, Purang Abolmaesumi, Teresa Tsang. University of British Columbia, Vancouver, BC, Canada.

Background: Unlike LV ejection fraction, which provides a valuable measure of systolic function, there is currently no analogous single measure of diastolic function. This study aimed to develop one using machine learning modeling of echocardiographic data. **Methods:** We screened consecutive echo studies performed at a tertiary care centre between February 1, 2010 and March 31, 2016. We excluded studies containing features that would interfere with diastolic function assessment as well as studies in which one or more diastology parameters were not reported. For the remaining studies, we graded diastolic function based on the 2016 American Society of Echocardiography (ASE) / European Association of Cardiovascular Imaging (EACVI) Guidelines and divided them into a training set (80%) and a validation set (20%), excluding indeterminate studies. We trained machine learning models (support vector machine, decision tree, XGBoost, and dense neural network) to classify studies by diastolic dysfunction severity based on the training set. Classification accuracy was assessed on the validation set. Transfer learning was used to generate a regression model from the dense neural network that outputs a continuous diastolic function score. **Results:** 28,986 studies were included. The models were able to classify studies with high agreement to ASE/EACVI guidelines (support vector machine 83% [panel A], decision tree 100% [panel B], XGBoost 100% [panel C], dense neural network 98% [panel D]). The continuous diastolic function score corresponded well with the ASE/EACVI guidelines, with mean scores of 1.00 ± 0.01 for studies with normal function; and 0.74 ± 0.05 , 0.51 ± 0.06 , and 0.27 ± 0.11 for mild, moderate, and severe diastolic dysfunction respectively (panel E). A score of <0.91 predicted abnormal diastolic function with an area under the receiver operator curve (AUROC) of 0.99 while a score of <0.65 predicted elevated filling pressure with an AUROC of 0.99. **Conclusion:** Machine learning models can assimilate echocardiographic data to

facilitate automatic classification according to ASE/EACVI guidelines as well as to generate a continuous diastolic function score.



PC-05

A Comparison of Artificial Intelligence-Enabled Global Longitudinal Strain Quantification to the Conventional On-Cart Analysis: A Single Center Experience
Pooja Prasad, Ahmed M. Altibi, Donna Kang, James Hodovan, Maros Ferencik, Lidija McGrath. OHSU, Portland, OR.

Background: Artificial intelligence (AI) provides an opportunity to increase efficiency and reduce inconsistencies in interpretation of global longitudinal strain (GLS). This study evaluated the performance of an automated, AI-enabled strain analysis as compared to the conventional on-cart strain analysis performed by cardiac sonographers. **Methods:** 100 transthoracic echocardiograms with on-cart GLS analysis performed between May and July 2021 were retrospectively identified. Included studies reported the GLS values of the apical 2-chamber, 3-chamber, and 4-chamber views, and were subsequently sent for analysis by the AI platform EchoGo Core (Ultrasonics Ltd, Oxford, UK). 98 studies were accepted for analysis by the AI software. Left ventricular ejection fraction (LVEF) obtained using the Simpson's biplane method was compared to LVEF obtained by AI. Individual and averaged GLS from the three apical views using on-cart analysis were compared to GLS values obtained from AI analysis. Linear regression and Bland-Altman plots were used to determine the correlation between the on-cart analysis and AI software. Regression models were used to adjust for age, gender, heart rate (HR), systolic blood pressure (SBP), and body mass index (BMI). **Results:** The mean patient age was 57.3 ± 13.7 years (53.5% female). Positive correlations were found between the on-cart and AI derived LVEF ($r=0.74$, $p<0.001$, n=89), and GLS values for the 2-chamber views ($r=0.59$, $p<0.001$, n=95), 3-chamber views ($r=0.67$, $p<0.001$, n=48), 4-chamber views ($r=0.65$, $p<0.001$, n=94), as well as averaged GLS from all 3 views ($r=0.77$, $p<0.001$, n=47). There was also a positive correlation between the averaged on-cart GLS (all 3 views) and the "best averaged" AI GLS (1, 2, or 3 averaged views; $r=0.71$, $p<0.001$, n=98). The correlation between the averaged on-cart and AI GLS remained significant after adjusting for age, gender, BMI, HR, and SBP ($r=0.73$, $p<0.001$). **Conclusion:** Strong positive correlations were found between LVEF and GLS values obtained by the conventional on-cart method and AI-enabled analysis. Our findings highlight the potential role for AI in LVEF and GLS assessment, thereby minimizing and eliminating inter-operator variabilities.